

19

D

Time for electron to travel in electric field, $t = \frac{s_x}{u}$

$$\text{At } t, \text{ displacement } s_y = \frac{1}{2}at^2 = \frac{1}{2}a\left(\frac{s_x}{u}\right)^2$$

Since a, u are constants, $s_y \propto s_x^2$

$$p = ks^2 = 1.0 \text{ mm}$$

Thus $q = k(2s)^2 = 4p = 4.0 \text{ mm}$ and so on.

